Munish Patwa

Data Scientist | ML Engineer | Data Analyst

Dortmund, Germany | +49 152 08243524 | Portfolio

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SUMMARY

M.Sc. Data Science student at TU Dortmund (available up to 20h/week). Built **RAG** pipelines, **multi-agent LLM** systems, end-to-end **ML forecasting** models, and **computer vision** applications using **Python**, **LangGraph**, **PyTorch**, and **Scikit-Learn**. Experienced with **Docker**, **CI/CD**, REST APIs, and model deployment. Seeking Werkstudent roles in Data Science, **ML** Engineering, or Data Analytics.

EDUCATION

M.Sc. Data Science | TU Dortmund — *Oct 2025 – Present*

Dortmund, Germany

Machine Learning, Statistics, Data Mining, Predictive Modelling, Deep Learning

B.Tech. Computer Science – Big Data Analytics | Ganpat University — *May 2021 – May 2025*

Mehsana, India

GPA: 9.13/10 | ML, Advanced Big Data Analytics, AI, DBMS, Cognitive Computing

WORK EXPERIENCE

AI/ML Engineer Intern | Ascion Global Partners — *Jan 2025 – May 2025*

Ahmedabad, India (Remote)

- Engineered a production **NLP** chatbot in **Python** handling **3** intent categories (bookings, queries, service requests), reducing average guest query resolution time by **~60%** based on response log benchmarks
- Improved **intent classification** F1-score by **~18%** by training and evaluating **Scikit-Learn** models across **5+** intent classes using a custom preprocessed conversational dataset
- Accelerated model iteration cycles by building a **Matplotlib**/**Plotly EDA** dashboard tracking chatbot usage patterns, error rates, and prediction confidence across test interactions
- Designed an end-to-end **ML** pipeline from raw data ingestion through **feature engineering**, model training, and evaluation, delivering results in structured reports to cross-functional stakeholders

Tech: Python, NLP, Scikit-Learn, Pandas, Matplotlib, Plotly, Intent Classification, ETL

Business Intelligence & Data Analytics Intern | A2Z Infra Solutions — *Jul 2024 – Dec 2024*

Ahmedabad, India (Hybrid)

- Built and maintained interactive **Power BI** dashboards across operations, project delivery, and finance, modelling data with **DAX** measures and **Power Query** (M) transformations from ERP and job-costing systems to give leadership a single source of truth
- Engineered documented **ETL** workflows that consolidated disparate spreadsheets and internal systems into a clean reportable schema, defining KPIs for job costing, inventory, scheduling, and procurement
- Surfaced cost-saving opportunities through exploratory data analysis, and safeguarded reporting accuracy with automated data validation, scheduled refreshes, and **row-level security** for governed multi-team access

Tech: Power BI, DAX, Power Query (M), ETL, Data Modelling, EDA, KPI Reporting, Row-Level Security

PROJECTS

Multi-Agent Research System: LangGraph, RAG & LLMs

https://multiagentresearchwithrag-munishpatwa.streamlit.app/ | https://github.com/Munish0303/Multi_Agent_Research_With_RAG

- Built and deployed a live **5**-agent autonomous research pipeline (**Orchestrator**, **Researcher**, **Analyst**, **Writer**, **Critic**) on **Streamlit Cloud** using **LangGraph** state machines with a **Critic**-driven critique-and-revise quality loop
- Engineered a dual-mode provider abstraction running fully offline on **Ollama** or in the cloud on **Groq**, auto-selecting the backend at runtime, with a **ChromaDB RAG** pipeline supporting PDF/TXT/MD ingestion and per-agent long-term memory
- Integrated multi-provider web search (**Tavily**/**Brave**/**DuckDuckGo** with backoff retry), **SQLite** run history, and PDF/Markdown report export; exposed the full pipeline as a **FastAPI REST API** with **async job polling**

Tech: Python, LangGraph, LLM, RAG, ChromaDB, Ollama, Groq, FastAPI, Streamlit, Multi-Agent, Generative AI

YouTube Trending Dashboard: Python, Pandas & Power BI

https://powerbiyoutubedashboard-munish.streamlit.app/ | https://github.com/Munish0303/POWERBI_Youtube_Dashboard

- Built an end-to-end analytics pipeline cleaning **342,656** trending records across **10** countries (from **375,942** raw rows) in **Python** (**Pandas**, ftfy), repairing Unicode mojibake and enforcing **0**-null QA assertions before export

- Engineered derived features (engagement rate, like/dislike ratio, days-to-trend, views buckets) and designed a **5**-page interactive **Power BI** dashboard covering executive KPIs, country comparison, category analysis, channels, and timing/virality
- Deployed a live **Streamlit** gallery sharing the dashboard pages; surfaced insights such as median **~1**-day time-to-trend and reach-vs-engagement divergence across content categories

Tech: Python, Pandas, ftfy, Power BI, Streamlit, EDA, Data Visualization, ELT, Data Cleaning

F1 Pit Stop Timing Classifier: XGBoost, SMOTE & SHAP

https://f1-pitstop-prediction-munish.streamlit.app/ | https://github.com/Munish0303/F1-Pitstop-Prediction

- Predicted whether an F1 driver pits on the next lap across **96,336** laps from the **2022**-**2024** seasons (OpenF1 API), achieving **ROC-AUC 0.979**, **PR-AUC 0.787** (vs. **0.032** baseline), and F1 **0.702** at a tuned threshold, beating **Logistic Regression** and **Random Forest** baselines
- Engineered **13** leakage-free features (tyre cliff proximity, degradation trend, circuit aggregates fitted on train only) and used a temporal **GroupShuffleSplit** on season; applied **SMOTE** on the **3.1%** minority class after the split to prevent cross-split contamination
- Improved methodological rigour by dropping a dominant but leaky feature (in-lap slow-lap flag), cutting its **52%** importance share with <**0.3% ROC-AUC** loss; explained predictions with **SHAP** TreeExplainer and shipped a modular **6**-step pipeline with batch inference

Tech: Python, XGBoost, SMOTE, SHAP, Scikit-Learn, Pandas, Forecasting, Feature Engineering, ETL

FireWatch: Real-Time Fire Detection & Alert System (Docker + CI/CD)

https://github.com/Munish0303/FireWatch

- Achieved real-time fire and smoke detection at <**200ms** inference latency by training **YOLOv8** on a custom **1,200**-image **Roboflow** dataset combined with **IoT** sensor fusion
- Containerized the full detection pipeline with **Docker**, deployed via **CI/CD** (**GitHub Actions**) on **Vercel** with **AWS S3** artifact storage, and automated multi-channel alerts via SMTP/**Twilio**

Tech: YOLOv8, Deep Learning, Computer Vision, Docker, CI/CD, AWS S3, IoT, Next.js, TypeScript, Vercel, REST APIs

Werkstudent Job Matcher: n8n, LLM & Arbeitsagentur API

https://github.com/Munish0303/n8n-jobsearcher

- Built a no-code **n8n** automation that ingests a CV via a hosted web form, parses the PDF, and scores live German job listings from the official Arbeitsagentur API **1**-**10** against the candidate profile using a **Groq** llama-**3.3**-**70**b **LLM**
- Automated end-to-end delivery: German-to-English title translation, role-weighted ranking, Excel report generation, and email dispatch, with **LLM**-call batching to stay within free API rate limits

Tech: n8n, LLM, Groq, REST API, Workflow Automation, PDF Parsing, Excel, SMTP

SKILLS

Programming: Python, SQL, R, TypeScript, JavaScript

ML / AI: Scikit-Learn, XGBoost, PyTorch, TensorFlow, Keras, Deep Learning, NLP, Computer Vision, SHAP, Forecasting, Feature Engineering, Predictive Modelling, Statistics

LLM & GenAI: LangGraph, LLM Orchestration, RAG, Multi-Agent Systems, Ollama, Groq, ChromaDB, Hugging Face, Prompt Engineering, Generative AI

Data & Analytics: Pandas, NumPy, Spark/PySpark, Power BI, Tableau (familiar), Matplotlib, Seaborn, Plotly, EDA, Data Cleaning, SMOTE, ETL, ELT, Data Pipelines, Probability Theory

MLOps & Cloud: Docker, CI/CD (GitHub Actions), MLOps, Kubernetes (familiar), AWS S3/EC2 (familiar), GCP Vertex AI (familiar), REST APIs, FastAPI, Vercel, Firebase, Git, GitHub

Databases: MySQL, PostgreSQL, Supabase

Web & Tools: React, Next.js, Streamlit, Microsoft Excel, Roboflow, Kaggle

Languages: English (C1), German (A2, improving), Gujarati (Native), Hindi (C1)